Magnetic confinement

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5.1 Pressure and density limits

pressure $p = 2nk_BT, \quad \beta = \dfrac{2\mu_0 p}{B^2} \quad \Rightarrow \quad p \le \beta_{\max} B^2/2\mu_0$

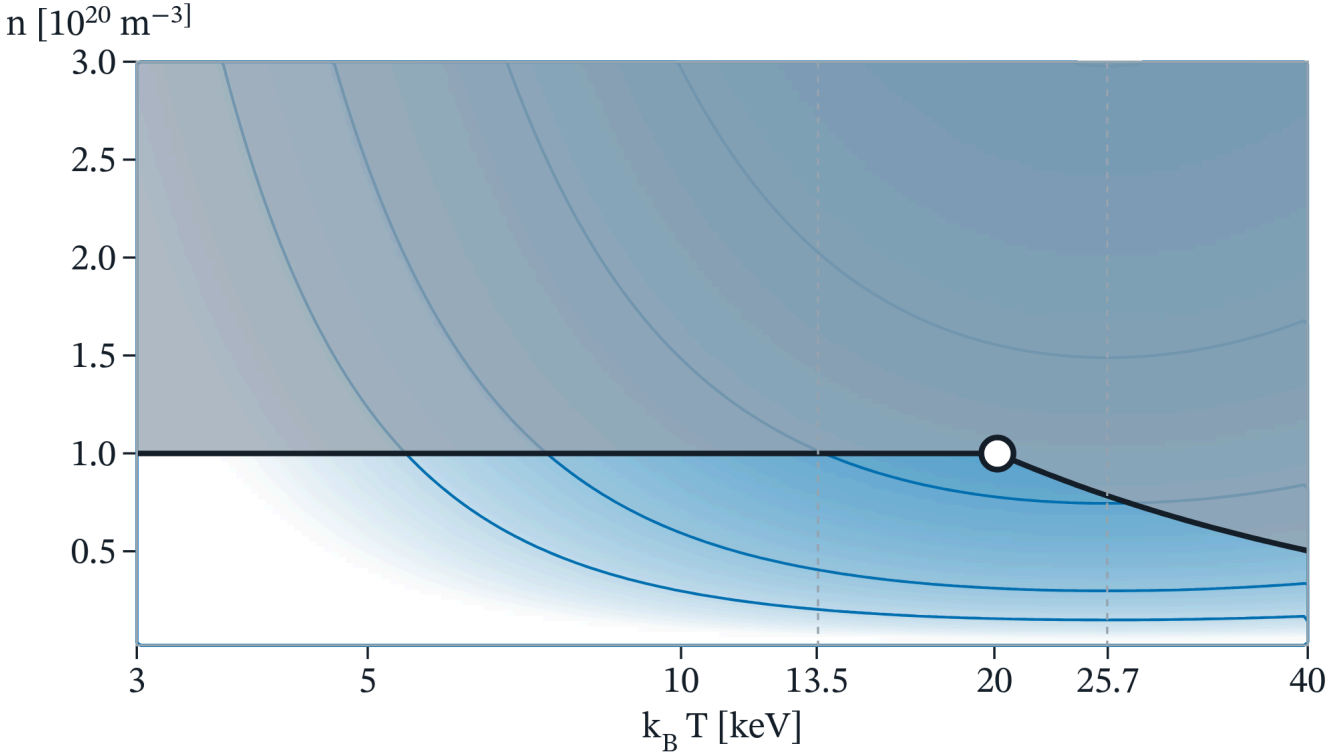
density $n \le n_G = I_p/(\pi a^2)$ $10^{20}\mathrm{m}^{-3}$, MA, m

$$p_{\text{fus}} \propto \frac{\langle \sigma v \rangle}{(k_B T)^2} \beta^2 B^4$$

fixed p: optimum 13.5 keV (4.4) twice B: 16 times p_{fus}

Greenwald · Plasma Phys. Control. Fusion 44, R27 (2002)

5.2 Density and pressure limits



τ_E needed [s]

0.3 1 3 10

blue lines: $\tau_E = 0.5, 1, 2, 5, 10$ s

grey: beyond density or β limit

dashed: 13.5 keV (best at fixed p), 25.7 keV (best at fixed n)

least τ_E needed ○

$k_B T = 20.1$ keV, $n = 1.00 \times 10^{20}$ m^{-3}

$\tau_E = 1.55$ s, $p = 6.4$ bar

limits cross at $T_\times = p_{max}/2n_{max} = 20.1$ keV

optimum = $T_\times$ clamped to [13.5, 25.7] keV

B [T]		6.0 T
β_{max} [%]		4.5 %
n_{max} [10^{20}]		1.00×10^{20} m^{-3}
target	○ Q = 1 ○ Q = 10 ◉ ignition	
no density limit (stellarator)	☐	

5.2 Limits and confinement scaling

limits $n \le n_G, \quad p \le \beta_{\max} B^2/2\mu_0 \quad \Rightarrow \quad T_\times = p_{\max}/2n_{\max}$

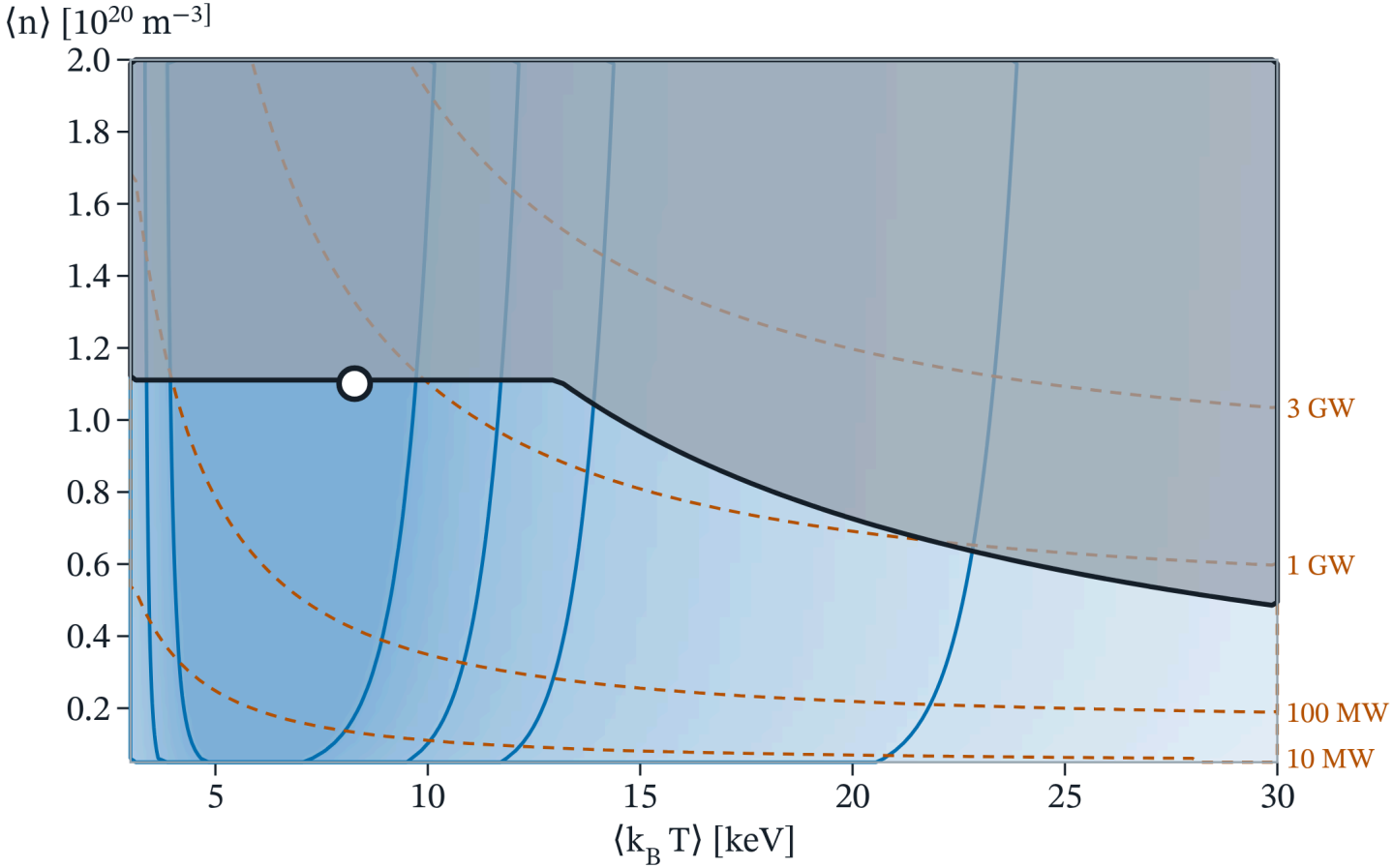
tokamak $T_\times \approx 0.39\,\beta_N a B/f_{\mathrm{GW}}$ keV ITER 8.8 keV

IPB98(y,2) $\tau_E \propto I^{0.93} R^{1.97} \overline{n}^{0.41} P^{-0.69} \ldots$

τ_E falls with heating power $\Rightarrow$ optimum $\approx$ 6–9 keV

ITER Physics Basis · Nucl. Fusion 39, 2175 (1999) · Yamada et al. · Nucl. Fusion 45, 1684 (2005)

5.2 Operating space of a device



gain Q

0.1 1 5 10 30

blue lines: $Q = 1, 5, 10, 30$ · orange: P_{fus} [MW]

grey: beyond Greenwald or β_N limit

ignited, highest P_{fus} ○

$k_B T = 8.3$ keV, $n = 1.10 \times 10^{20}$ m^{-3}

$Q = \infty$, $P_{fus} = 688$ MW

$\tau_E = 3.05$ s, $\beta = 2.7$ %

$P_{ext} = 0$ MW, $P_\alpha = 139$ MW

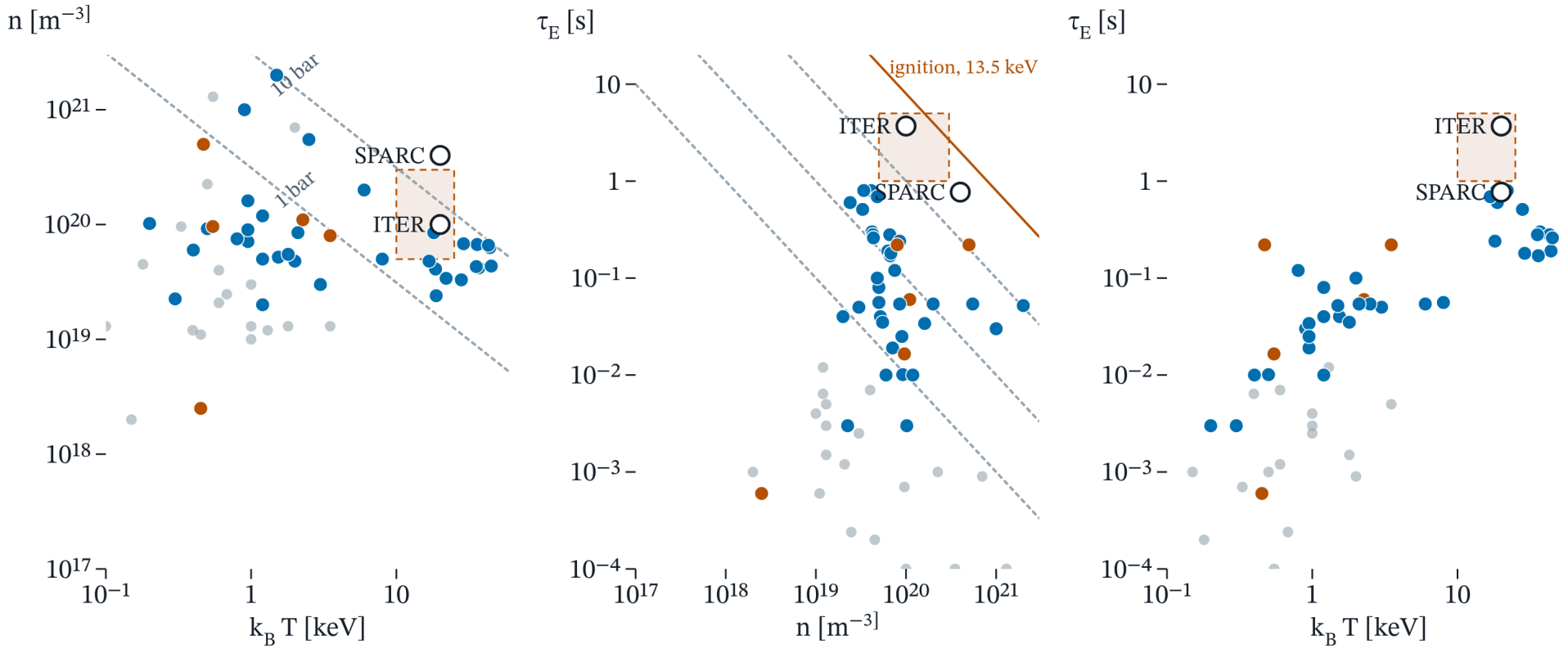
0-D, parabolic profiles ($\alpha_n = 0.2$, $\alpha_T = 1$), fuel 85 % of n_e, axes volume averages, $T_e = T_i$, bremsstrahlung only; τ_E = H·IPB98(y,2) or H·ISS04 with $P = W/\tau_E$ solved self-consistently. Trends, not design values.

device ITER (tokamak)

B [T] 5.3 T

H factor 1.00

5.2 Where present machines stand



● tokamak ● stellarator ● other magnetic ○ design ▭ D–T reactor window

Peak values; gain curves for peaked profiles, Z_{eff} 1.5–3.4; data: Wurzel & Hsu, Phys. Plasmas 29, 062103 (2022)

view ◉ three cross-sections ○ triple product